CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 5: Quadratic Equations

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The quadratic formula

1. Solve $2x^2 - 5x - 4 = 0$, giving the answers correct to three significant figures. ICSE 2025, 4 marks
2. Solve $2x^2 - 10x + 5 = 0$, giving the answers correct to three significant figures. ICSE 2024, 3 marks
3. Solve $x^2 + 4x - 8 = 0$, giving the answers correct to one decimal place. ICSE 2023, 4 marks
4. Solve $x^2 - 7x + 3 = 0$, giving the answers correct to two decimal places. ICSE 2020, 3 marks
5. Solve $x^2 - 4x - 8 = 0$, giving the answers correct to three significant figures. ICSE 2019, 3 marks
6. Solve $x^2 + 7x = 7$, giving the answers correct to two decimal places. ICSE 2018, 4 marks
7. Solve $4x^2 - 5x - 3 = 0$, giving the answers correct to two decimal places. ICSE 2017, 4 marks

2 Nature of the roots

8. The equation $3x^2 + \sqrt{7}\,x + 2 = 0$ has: (a) two equal real roots (b) two distinct real roots (c) more than two real roots (d) no real roots ICSE 2025, 1 mark
9. The roots of $px^2 - qx + r = 0$ are real and equal when: (a) $p^2 = 4qr$ (b) $q^2 = 4pr$ (c) $-q^2 = 4pr$ (d) $p^2 > 4qr$ ICSE 2024, 1 mark
10. Find the values of k for which $x^2 + 4kx + (k^2 - k + 2) = 0$ has equal roots. ICSE 2018, 3 marks

3 Standard form and roots

11. 3 is a root of $x^2 - px + 3 = 0$. Find p. (a) 4 (b) 3 (c) 5 (d) 2 ICSE 2023, 1 mark

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